

Note

Distributions Modulo Subgroups of $GF(q)^*$

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The main purpose of this paper is to determine the number of symbols in each equivalence class modulo subgroups of $GF(q)^*$ for a certain class of linear recursive sequences with irreducible generating polynomial. The method used is to map sequences onto maximal linear recursive sequences.

1. INTRODUCTION

We consider recursive sequences over $GF(q)$ (the Galois field with q elements) generated by an irreducible polynomial f of degree n . All roots of f have the same order e (called the exponent of f) in the multiplicative group $GF(q^n)^* = GF(q^n) - \{0\}$. Every nonzero recursive sequence generated by f is periodic with period e , and there are (up to translation) $(q^n - 1)/e$ nonzero f -sequences [1], [2].

If $e = q^n - 1$ the polynomial f is primitive, and in a single period of the nonzero f -sequence zeros appear $q^{n-1} - 1$ times and every nonzero element appears q^{n-1} times. This is easily seen because every n -tuple except all zero must appear exactly once in a period.

If $e < q^n - 1$ the distribution of elements within each individual nonzero f -sequence need not be so nearly uniform.

We shall show that for certain values of e the distribution of each f -sequence is uniform over the cosets of a certain subgroup of the multiplicative group of $GF(q)^*$. Specifically this occurs when $[e, q - 1]$, the least common multiple of e and $q - 1$, is $q^n - 1$.

2. DISTRIBUTION FORMULAS

If f is an irreducible polynomial of degree n over $GF(q)$, f has a factorization $f(x) = \prod_{i=1}^n (x - f_i)$ with f_i in the splitting field $GF(q^n)$ of f . Let $\{a_n\}$ be a f -sequence and b any element in $GF(q)^* = GF(q) - \{0\}$, then the

sequence $\{a_n b^n\}$ is a $\prod_{i=1}^n (x - f_i b)$ -sequence. [1, p. 92]. When $\{a_n b^n\}$ is a maximal recurring sequence of period $q^n - 1$ its known distribution of elements from $GF(q)$ can be utilized to evaluate the distribution within $\{a_n\}$. A lemma will be needed in order to determine which sequences $\{a_n\}$ can be mapped onto maximal sequences by appropriate choice of an element b in $GF(q)^*$.

LEMMA 1. *If B is a subgroup of a cyclic group G and a is an element in G such that $[\mid B \mid, \mid a \mid] = \mid G \mid$, then there exists $b \in B$ such that ab is a generator of G . Furthermore, b can be chosen so that, $\mid b \mid$ the order of b , is the least divisor d of $\mid B \mid$ such that $[\mid a \mid, d] = \mid G \mid$.*

Proof. Let $b \in B$ be any element of order d where d is the least divisor of $\mid B \mid$ (the order of B) such that $[\mid a \mid, d] = \mid G \mid$. If y is a generator in G , there exist integers s and t such that

$$\begin{aligned} a &= y^{(\mid G \mid / \mid a \mid)s} & \text{with } (\mid a \mid, s) &= 1 & \text{and} \\ b &= y^{(\mid G \mid / \mid b \mid)t} & \text{with } (\mid b \mid, t) &= 1. \end{aligned}$$

It is easily seen that

$$ab = y^{(s\mid b \mid + t\mid a \mid)(\mid a \mid, \mid b \mid)^{-1}}$$

since $[\mid a \mid, \mid b \mid] = \mid G \mid$. Hence ab is a generator of G if and only if $(s\mid b \mid + t\mid a \mid)(\mid a \mid, \mid b \mid)^{-1}$ is prime to $\mid G \mid$.

Suppose p is a prime dividing $\mid G \mid$. Then p does not divide $\mid a \mid$ and $\mid b \mid$ to the same power.

Without loss we may assume that there exists an integer i such that $p^i \mid \mid a \mid$ and $p^i \nmid \mid b \mid$. Then p is a divisor of $\mid a \mid(\mid a \mid, \mid b \mid)^{-1}$ but not of $\mid b \mid(\mid a \mid, \mid b \mid)^{-1}$. Since $(\mid a \mid, s) = 1$ p is not a divisor of s and thus $p \times (s\mid b \mid + t\mid a \mid)(\mid a \mid, \mid b \mid)^{-1}$. This proves that ab is a generator in G . ■

To every irreducible polynomial f over $GF(q)$ of degree n and exponent e such that $[e, q - 1] = q^n - 1$ we associate the subgroup $G(f)$ of $GF(q)^*$ generated by the elements of order d where d is the least divisor of $q - 1$ such that $[d, e] = q^n - 1$.

THEOREM 1. *Let f be an irreducible polynomial over $GF(q)$ of degree n and exponent e with $[e, q - 1] = q^n - 1$. Suppose $\{a_n\} \neq \{0\}$ is a recurrence sequence with generating polynomial f , then the frequency of zeros in $\{a_n\}$ is $(q^{n-1} - 1)(q^n - 1)^{-1}$ and the frequency of elements from every coset of $G(f)$ in $GF(q)^*$ is $q^{n-1}(q^n - 1)^{-1} \mid G(f) \mid$ where $\mid G(f) \mid$ is the number of elements in $G(f)$.*

Proof. Let a be a root in f and b any generator of $G(f)$. From the lemma we obtain that ab is a generator of $GF(q^n)^*$. If $\{a_n\}$ is a nonzero f -sequence, the sequence $\{a_nb^n\}$ is a maximal recurring sequence of period $q^n - 1$. In $\{a_nb^n\}$ each n -dimensional vector except $\vec{0}$ occurs once in each period. Since $a_nb^n = 0$ implies $a_n = 0$ it follows that the frequency of zeros in $\{a_n\}$ is equal to that in $\{a_nb^n\}$ which is $(q^{n-1} - 1)(q^n - 1)^{-1}$. For any $a_n \in GF(q)^*$, a_n and a_nb^n will belong to the same coset of $G(f)$ in $GF(q)^*$, and it follows that the frequency of elements from each coset of $G(f)$ in $\{a_n\}$ is equal to that in $\{a_nb^n\}$ which is $q^{n-1}(q^n - 1)^{-1} |G(f)|$. ■

In particular the number of nonzero elements in every f -sequence (except 0) is equal. i.e. The cyclic code of block-length e consisting of all f -sequences is an equidistant code.

In the remaining part of the paper we will give explicit values of the exponents e such that $[e, q - 1] = q^n - 1$ as a function of q and n and calculate the order of $G(f)$. Lemma 2, whose simple proof is omitted, will be needed in order to prove Theorem 2.

LEMMA 2. $(q - 1, (q^n - 1)(q - 1)^{-1}) = (q - 1, n)$.

THEOREM 2. Let f be an irreducible polynomial of degree n , and exponent e . Then $[e, q - 1] = q^n - 1$ if and only if $e = (q^n - 1)/s$ where s divides $q - 1$ but is relatively prime to n . In this case, $|G(f)| = (q - 1)/u$ where u is the largest divisor of $q - 1$ relatively prime to s .

Proof. Let e be a divisor of $q^n - 1$. By applying Lemma 2 one gets

$$\begin{aligned} [e, q - 1] &= \left[\frac{q^n - 1}{q - 1} \cdot \frac{q - 1}{s}, q - 1 \right] \\ &= (q^n - 1) \frac{q - 1}{s} \left(n \frac{q - 1}{s}, q - 1 \right)^{-1} \end{aligned}$$

which is equal to $q^n - 1$ if and only if s is prime to n . The last part of the theorem is immediate from the construction of $G(f)$. ■

EXAMPLE. Let $q = 64$ and $n = 3$. Then $s/q - 1$ and $(s, n) = 1$ if and only if $s = 1$ or $s = 7$. For $s = 7$ we have $e = 3^3 \cdot 19 \cdot 73$ and $q - 1 = 3^2 \cdot 7$. The order of $G(f)$ is the product of the prime powers in $q - 1$ where the exponent is greater than the exponent in the expansion of e . In this example $|G(f)| = 7$.

REFERENCES

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